Alcohols and Ethers

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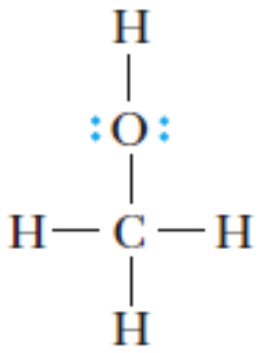
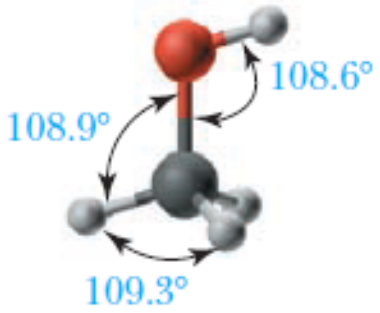
Alcohols

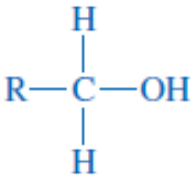
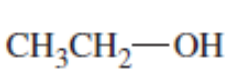
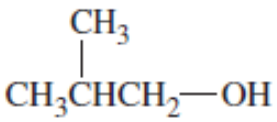
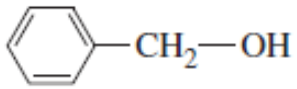
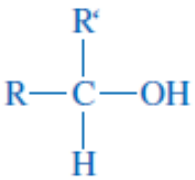
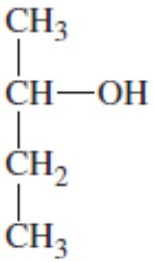
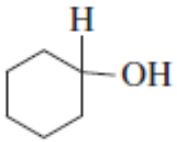
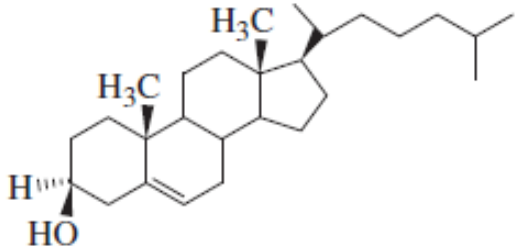
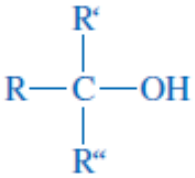
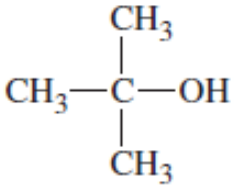
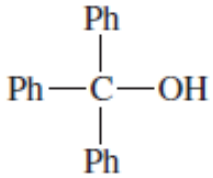
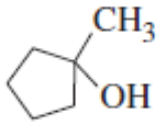
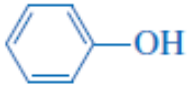
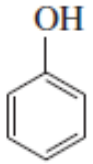
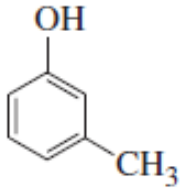
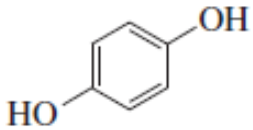
- The functional group of an alcohol is an hydroxy group (OH) bonded to an sp^3 hybridised carbon atom.
- The oxygen atom of OH is also sp^3 hybridised. Two of the sp^3 hybrid orbitals of oxygen form sigma bonds to carbon and hydrogen, and the other two sp^3 hybrid orbitals hold the unshared pair of electrons.





Classification of Alcohols

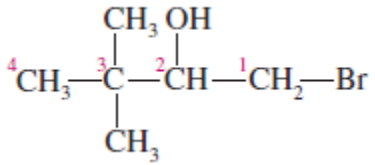
- One way of organising alcohols is to classify them based on their **carbinol carbon** atom – the carbon bonded to the OH.
- If this carbon is primary (bonded to one other carbon atom), the alcohol is classified as a primary alcohol.
- A secondary alcohol's OH group is bonded to a carbon atom that is bonded to two other carbon atoms.
- Compounds in which a hydroxy group is directly bonded to a benzene ring are called phenols.

Type	*Structure*	*Examples*		
Primary alcohol		 ethanol	 2-methylpropan-1-ol	 benzyl alcohol
Secondary alcohol		 butan-2-ol	 cyclohexanol	 cholesterol
Tertiary alcohol		 2-methylpropan-2-ol	 triphenylmethanol	 1-methylcyclopentanol
Phenols		 phenol	 3-methylphenol	 hydroquinone

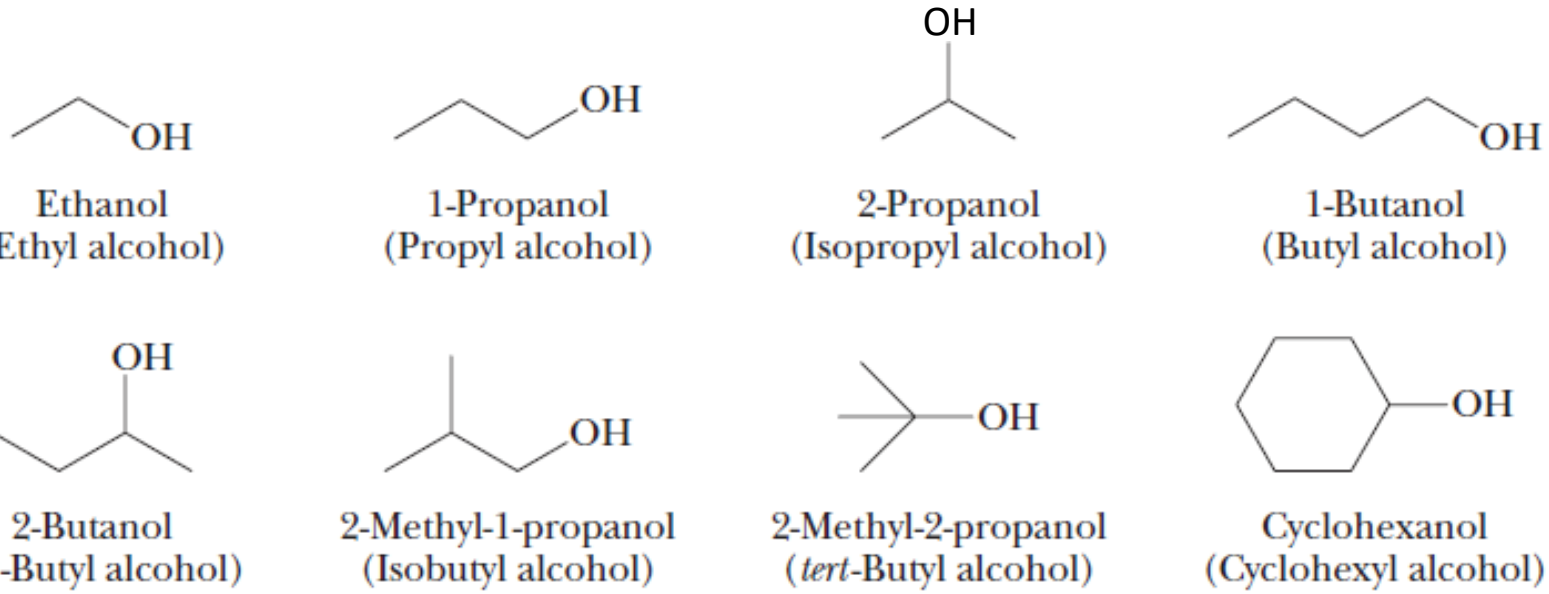
Nomenclature of Alcohols

- We derive the IUPAC name for alcohols in a similar way to that of alkanes, except that the ending of the parent chain changes from *–e* to *–ol*.

1. Select the longest chain of carbon atoms that contains the OH group, and number that chain from the end closest to OH. In naming alcohols, the OH group takes precedence over alkyl groups and halogens.
2. Name the parent chain based on the number of carbon atoms. Drop the final *–e* from the alkane name and add the suffix *–ol* to give the root name.
3. Use a number to show the location of the OH group. For cyclic alcohols, numbering starts at the carbon atom bonded to the OH.
4. Name and number substituents, and list them in alphabetical order.



1-bromo-3,3-dimethylbutan-2-ol

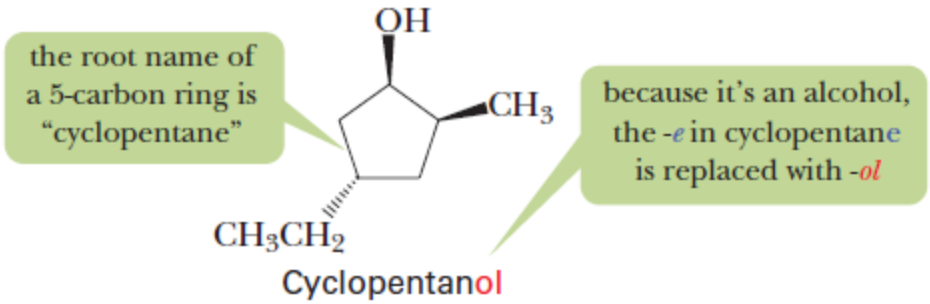


IUPAC Name Common Name	Ethanol (Ethyl alcohol)	1-Propanol (Propyl alcohol)	2-Propanol (Isopropyl alcohol)	1-Butanol (Butyl alcohol)
IUPAC Name Common Name	2-Butanol (*sec*-Butyl alcohol)	2-Methyl-1-propanol (Isobutyl alcohol)	2-Methyl-2-propanol (*tert*-Butyl alcohol)	Cyclohexanol (Cyclohexyl alcohol)

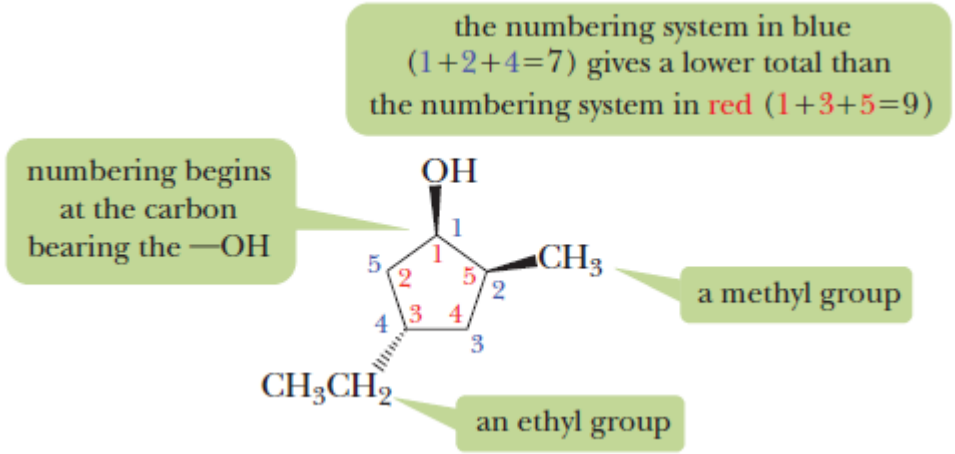
Common names are given in brackets

- To name *cyclic alcohols*:

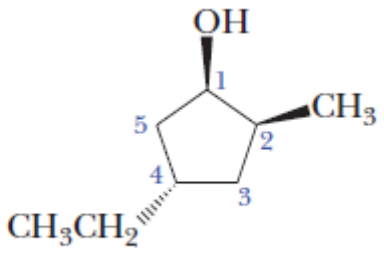
1. First determine the root name of the cycloalkane, replacing *–e* with *–ol*.



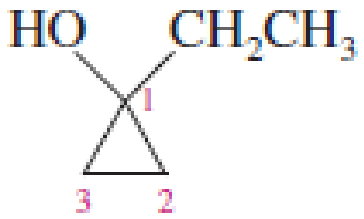
2. Name and number the substituents. Numbering begins at the carbon bearing –OH group, and proceeds in the direction that gives the lowest total for all substituents.



3. Name substituents in alphabetical order, preceded by its position on the ring. Position 1 for the OH group is assumed.

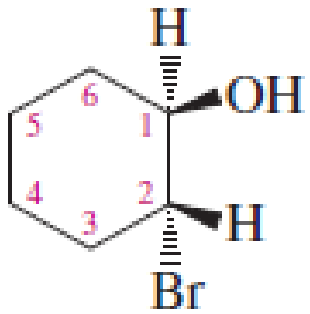


4-Ethyl-2-methylcyclopentanol



1-ethylcyclopropanol

Don’t forget to indicate stereochemistry!

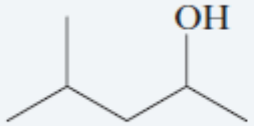


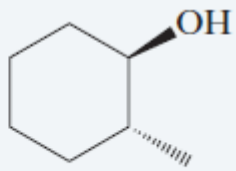
trans-2-bromocyclohexanol

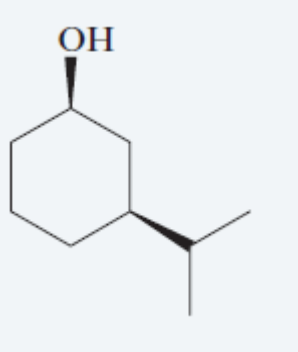
- Practice exercise:

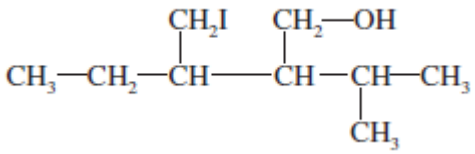
Write the IUPAC name for each of the following alcohols:

(a) $CH_3(CH_2)_6CH_2OH$

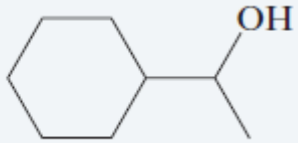
(b) 

(c) 

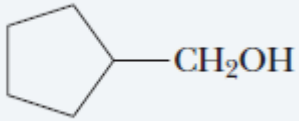




Classify each of the following alcohols as primary, secondary or tertiary:

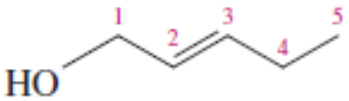
(a) 

(b) CH_3
CH_3COH
CH_3

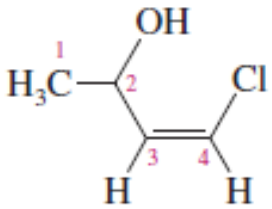
(c) 

Unsaturated alcohols

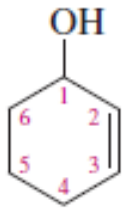
- Compounds that contain an OH group and a double or triple bond are referred to as *unsaturated alcohols*. To name an unsaturated alcohol:
- The alcohol takes precedence over double or triple bonds, therefore use the *–ol* suffix after the alkene or alkyne name.
- Number the parent chain so as to give the OH the lowest number.
- Show the double bond by changing the infix of the parent alkane from *–an* to *–en* for alkenes, and –an to –yne for alkynes, and the alcohol by changing the suffix from *–e* to *–ol*.
- Use numbers to show the location of the double bond/triple bond and the hydroxy group.



trans-2-penten-1-ol
trans-pent-2-en-1-ol

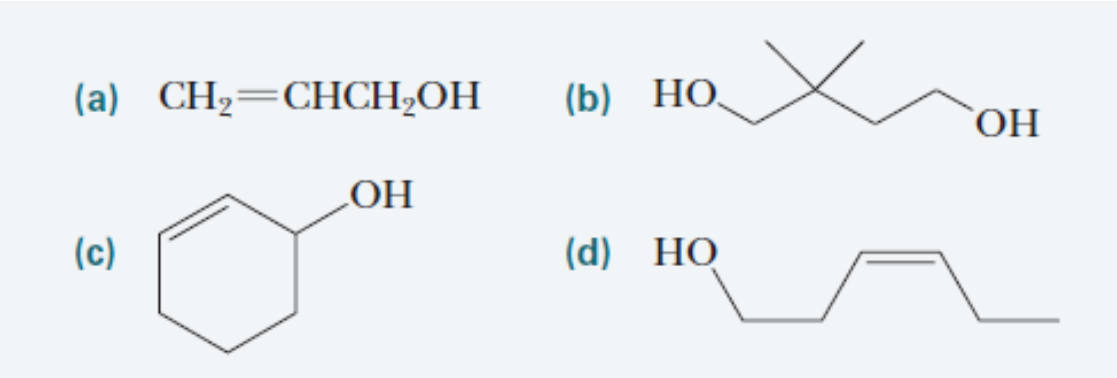


(Z)-4-chloro-3-buten-2-ol
(Z)-4-chlorobut-3-en-2-ol



2-cyclohexen-1-ol
cyclohex-2-en-1-ol

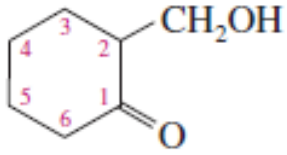
Practice exercise: Write the IUPAC names for each of the following alcohols:



- The –OH group is named as a **hydroxy** substituent when it appears on a structure with a higher priority functional group, or when the compound is too complicated to name as an alcohol.

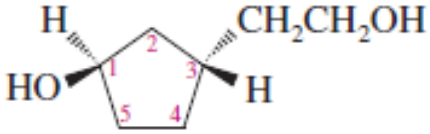
Priority of functional groups in naming organic compounds

acids (highest)
esters
aldehydes
ketones
alcohols
amines
alkenes, alkynes
alkanes
ethers
halides (lowest)

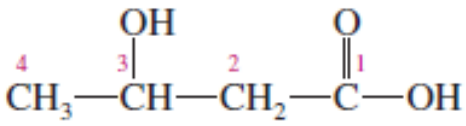


2-(hydroxymethyl)cyclohexanone

Ketone – higher functional priority



trans-3-(2-hydroxyethyl)cyclopentanol



3-hydroxybutanoic acid

Carboxylic acid – higher functional priority

Physical Properties of Alcohols

- The most important physical property of alcohols is the polarity of their –OH groups.
- Large difference in electronegativity between oxygen and carbon (3.5 – 2.5 = 1.0) and between oxygen and hydrogen (3.5 – 2.1 = 1.4).
- Therefore both C-O and O-H bonds are polar covalent bonds.
- Alcohols have higher boiling points than alkanes of similar molecular weight due to the presence of hydrogen bonding between the molecules, which are stronger than dispersion forces. Thus require more energy to break.

- As the number of hydrogen bonding per molecule increases, the boiling point also increases. For example 1-pentanol (138 °C) and 1,4-butanediol (230 °C), both of which have approximately the same molecular weight.
- As molecular weight of alcohols increase, boiling points increase due to increased strength of dispersion forces. This applies to all types of compounds.
- Methanol and ethanol are free-flowing, volatile liquids that have fruity odours.
- The higher alcohols (butanol to decanol) are viscous liquids, and some of the highly branched isomers are solids at room temperature.

- Alcohols are much more soluble in water than alkanes, alkenes, or alkynes of comparable molecular weight. Due to the presence of hydrogen bonding between the alcohols and water molecules.
- Low-molecular weight alcohols are completely soluble (miscible) in water.
- Similarly, alcohols are much better solvents for polar compounds than are hydrocarbons.
- The hydroxy groups is said to be **hydrophilic** (water-loving) due to its affinity to water, and the alkyl chain is said to be **hydrophobic** (water-hating).
- As the alcohol molecular weight increases, solubility in water decreases due to the increase in size of the hydrocarbon (non-polar) portion of the chain.

Boiling points and solubilities in water of five groups of alcohols and alkanes of similar molecular weight

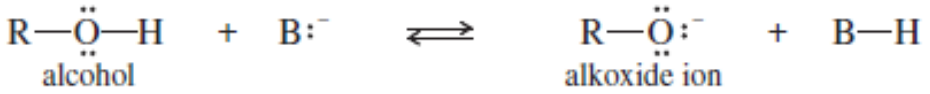
Structural Formula	Name	Molecular Weight	Boiling Point (°C)	Solubility in Water
CH_3OH	methanol	32	65	infinite
CH_3CH_3	ethane	30	−89	insoluble
CH_3CH_2OH	ethanol	46	78	infinite
$CH_3CH_2CH_3$	propane	44	−42	insoluble
$CH_3CH_2CH_2OH$	1-propanol	60	97	infinite
$CH_3CH_2CH_2CH_3$	butane	58	0	insoluble
$CH_3CH_2CH_2CH_2OH$	1-butanol	74	117	8 g/100 g
$CH_3CH_2CH_2CH_2CH_3$	pentane	72	36	insoluble
$CH_3CH_2CH_2CH_2CH_2OH$	1-pentanol	88	138	2.3 g/100 g
$HOCH_2CH_2CH_2CH_2OH$	1,4-butanediol	90	230	infinite
$CH_3CH_2CH_2CH_2CH_2CH_3$	hexane	86	69	insoluble

Uses of Alcohols

- Alcohols are among the most important chemical commodities worldwide.
- The most commonly manufactured alcohols include methanol, ethanol, isopropyl alcohol (propan-2-ol), and ethylene glycol (ethane-1,2-diol).
- Methanol is commonly used as a solvent for organic reactions. It is also a good fuel for internal combustion engines.
- Like methanol, ethanol is commonly used as a solvent for laboratory use. It is the alcohol in alcoholic beverages.
- Propan-2-ol (isopropyl alcohol) is used as a disinfectant

Chemical Properties of alcohols

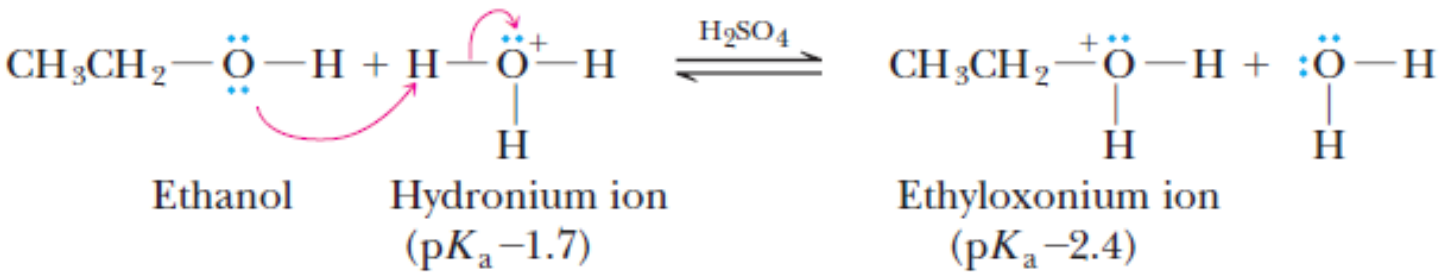
- The hydroxy proton of alcohols is weakly acidic. A strong base can easily deprotonate the alcohol to give an **alkoxide** ion.



Example

- Alkoxides are strong nucleophiles and strong bases.

- In the presence of strong acids, the oxygen atom of alcohols functions as a weak base and reacts with an acid by proton transfer to form an oxonium ion.

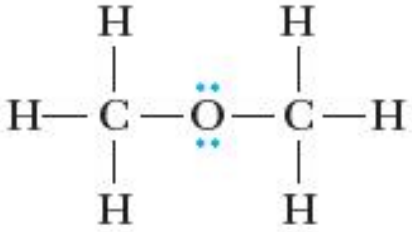


- Thus, alcohols can act as both acids and bases.

Ethers

Ethers

- The functional group of an ether is an oxygen atom covalently bonded to two carbon atoms.
- Dimethyl ether (CH_3OCH_3) is the simplest ether. Two of the sp^3 hybrid orbitals on the oxygen form sigma bonds to the two carbon atoms. The other two sp^3 hybrid orbitals on oxygen hold the unshared pair of electrons.

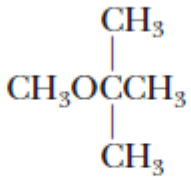


IUPAC Nomenclature

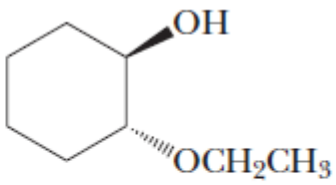
- In the IUPAC system, ethers are named by selecting the longest carbon chain as the parent chain, and naming the –OR group bonded to it as an ***alkoxy*** group.
- Common names are derived by listing the alkyl groups bonded to oxygen in alphabetical order and adding the word *ether*.
- Common names are commonly used for low-molecular weight ethers.

$CH_3CH_2OCH_2CH_3$

Ethoxyethane
(Diethyl ether)

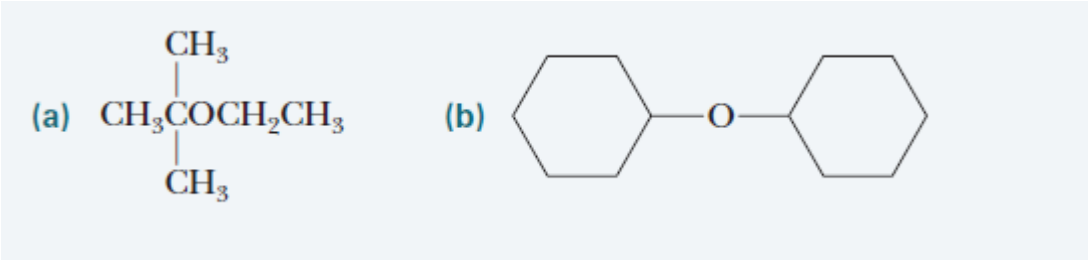


2-Methoxy-2-methylpropane
(methyl *tert*-butyl ether, MTBE)



(1*R*,2*R*)-2-Ethoxycyclohexanol
(*trans*-2-Ethoxycyclohexanol)

- Practice exercise 1: Write the IUPAC name for the following compounds:



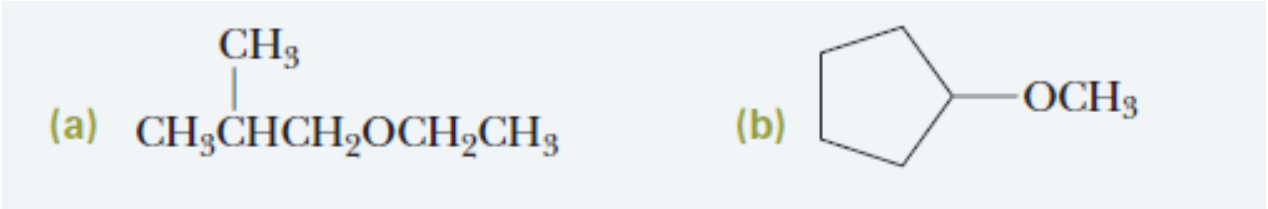
Priority of functional groups in naming organic compounds

acids (highest)
esters
aldehydes
ketones
alcohols
amines
alkenes, alkynes
alkanes
ethers
halides (lowest)

Ans: a) 2-ethoxy-2-methylpropane

b) cyclohexoxycyclohexane

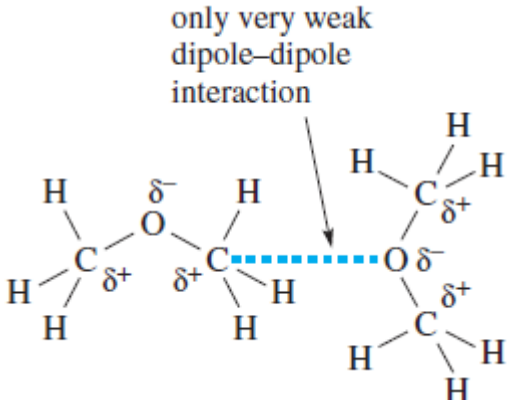
Practice exercise 2: Write the IUPAC name for the following compounds:

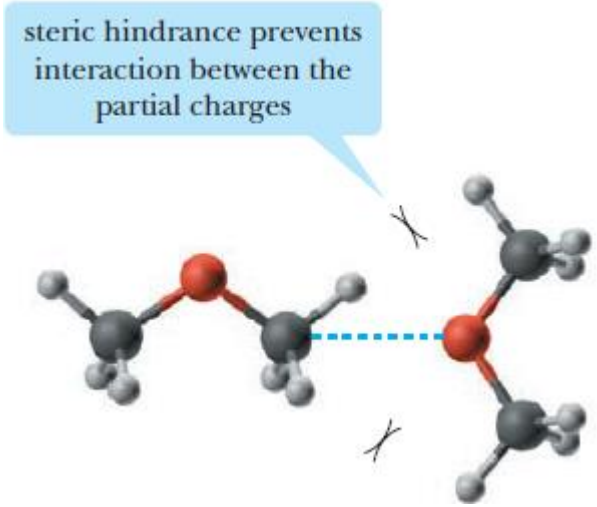


- Ethers are relatively unreactive and they are rarely used as synthetic intermediates.
- Because of their stability, they are usually used as solvents for organic reactions.
- Diethyl ether is used as a volatile starting fuel for diesel and gasoline engines. It was used in the past as an anaesthetic for surgical operations, but this was discontinued to side effects and its flammability.

Physical Properties of Ethers

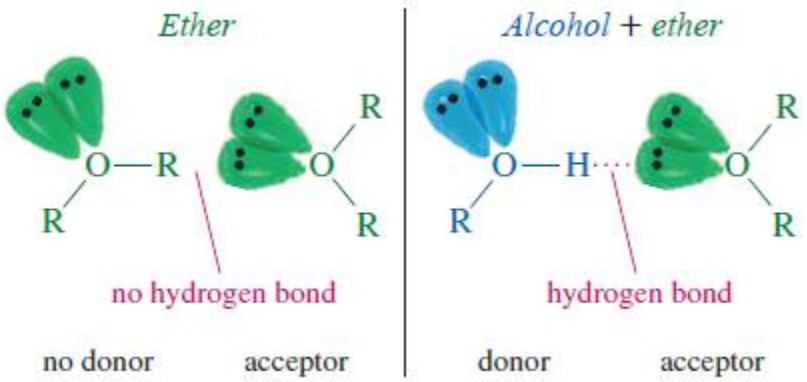
- Ethers are polar molecules.
- Because of steric hindrance, however, weak forces of attraction exist between ether molecules in the pure liquid. Therefore they have lower boiling points than alcohols of similar molecular weight.
- Boiling points of ethers are similar to those of hydrocarbons of comparable molecular weight.



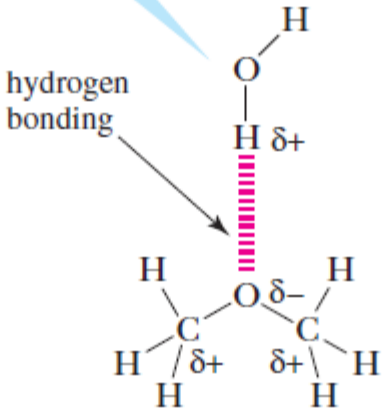


- For example, ethanol has a higher boiling point than dimethyl ether due to the ability of ethanol molecules to hydrogen bond with each other. This is not possible between ether molecules.

- Due to the partially negatively charged oxygen of ethers, they are able to hydrogen bond with water (but not with themselves!). Thus, ethers are more soluble in water than are hydrocarbons of similar molecular weight.



dimethyl ether in water. The partially negative oxygen of the ether is the hydrogen-bond acceptor, and a partially positive hydrogen of a water molecule is the hydrogen-bond donor



- Practice exercise: Arrange the following compounds in order of increasing solubility in water:

$CH_3OCH_2CH_2OCH_3$	$CH_3CH_2OCH_2CH_3$	$CH_3CH_2CH_2CH_2CH_2CH_3$
Ethylene glycol dimethyl ether	Diethyl ether	Hexane